Decision Trees

Decision Trees

- What is a Decision Tree?
- Classification is a two-step process in machine learning
  - learning step and
  - prediction step,
- In the learning step, the model is developed based on given training data.
- In the prediction step, the model is used to predict the response for given data.
- Decision Tree is one of the popular classification algorithm - supervised and non-parametric
- It can be used for solving regression and classification problems

Decision Trees

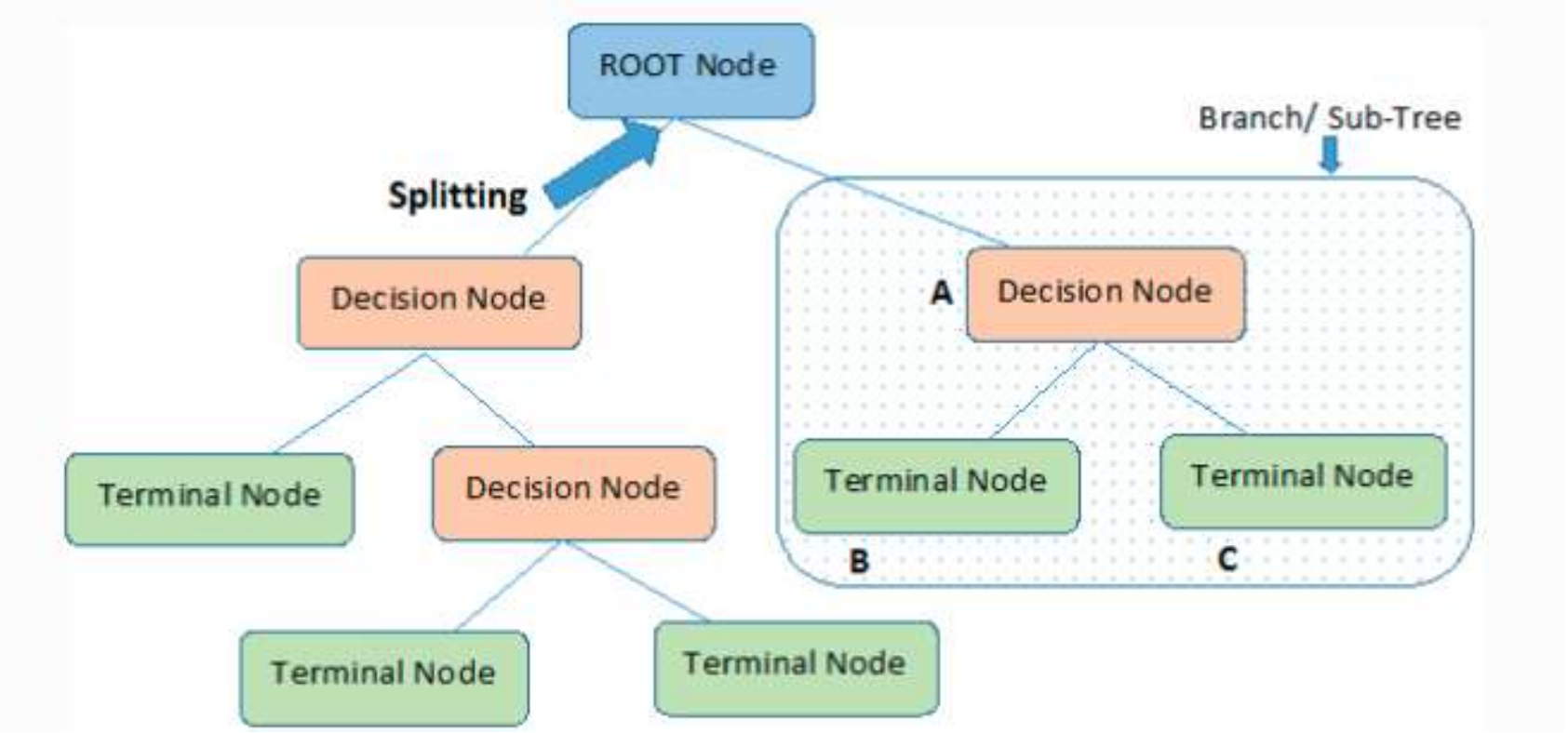
- ID3 → (extension of D3), uses entropy for classification
- C4.5 → (successor of ID3), uses gain ratio for classification
- C5 → (successor of C4.5), efficient than C4.5
- CART → (Classification And Regression Tree)
- Decision Trees are an important type of algorithm for predictive modeling in machine learning
  - The classical decision tree algorithms – ID3 (Iterative Dichotomiser 3), C4.5, CART (Classification and Regression Trees)
  - Modern variations - random forest

  Meaning
  - verb (used with object),di·chot·o·mized, di·chot·o·miz·ing.
  - **to divide or separate into two parts**, kinds, etc

Structure of Decision Trees

1. The root node
2. The Decision node
3. Leaf/Terminal node

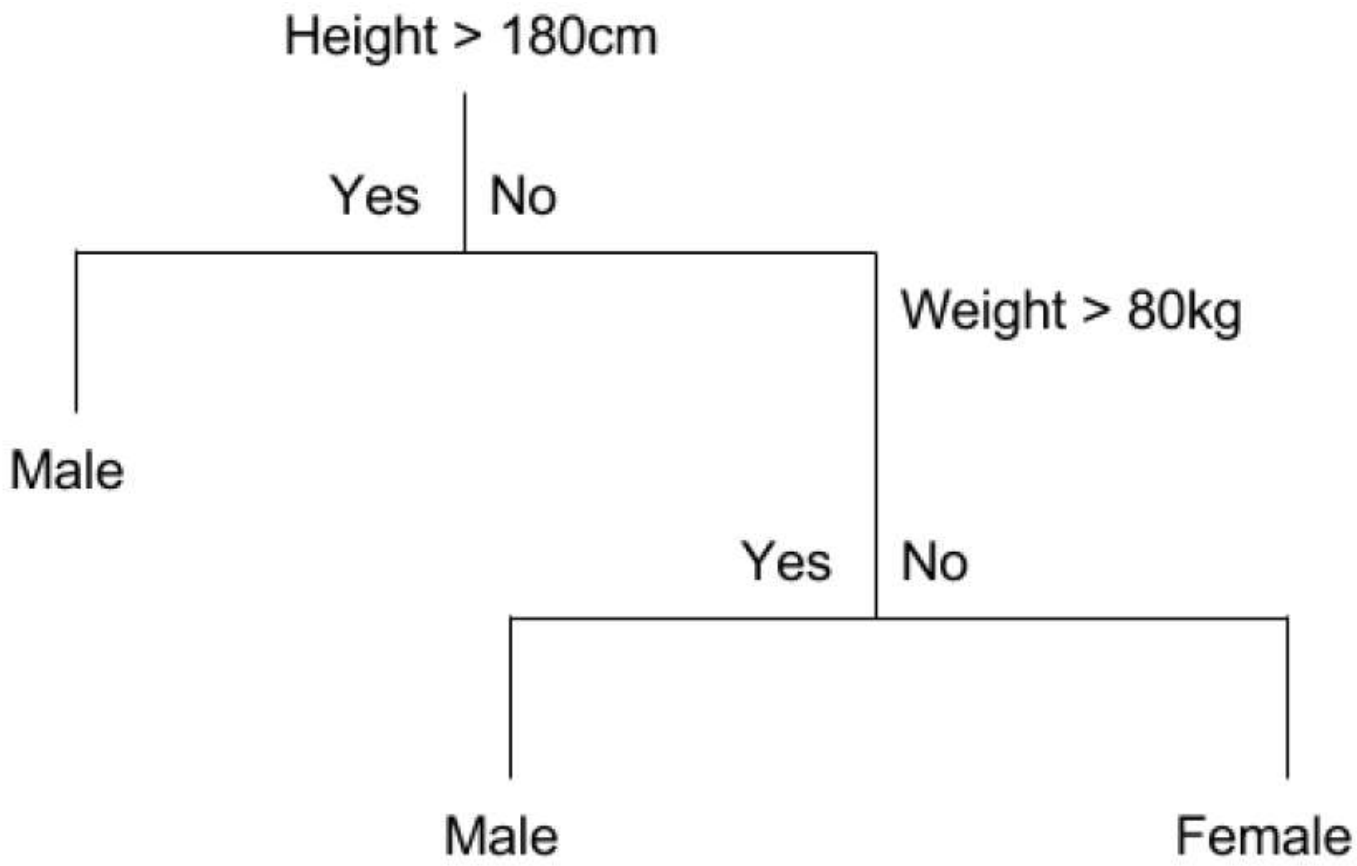
Q: What is Pruning?



Decision Trees - Terminology

- Root Node: It represents the entire population or sample and this further gets divided into two or more homogeneous sets
- Splitting: It is a process of dividing a node into two or more sub-nodes
- Decision Node: When a sub-node splits into further sub-nodes, then it is called the decision node
- Leaf / Terminal Node: Nodes do not split is called Leaf or Terminal node
- Pruning: When we remove sub-nodes of a decision node, this process is called pruning, the opposite process is splitting
- Branch / Sub-Tree: A subsection of the entire tree is called branch or sub-tree
- Parent and Child Node: A node, which is divided into sub-nodes is called a parent node of sub-nodes whereas sub-nodes are the child of a parent node

Example Decision Tree



When can we stop building the decision tree?

Decision Trees – advantages

- Easy to model and interpret
- Simple to understand
- The input and output attributes can be discrete or continuous
- There can be a high degree of nonlinearity in the relationship between target variables and predictor(input) variables
- We can quickly train a decision tree

Decision Trees – disadvantages

- It’s difficult to determine how deeply a decision tree can be grown/ when to stop
- If the training dataset has errors or missing values, then the decision tree constructed may become unstable or biased
- When the training data has continuous values, handling it can become computationally complex, and therefore it has to be discretized again
- A complex decision tree can result in overfitting issues
- Decision trees are not well suited when we are classifying multiple output classes
- Learning an optimal decision tree is known as NP complete problem

Example 6.2: Predict a student's academic performance of whether he will pass or fail based on the given information such as 'Assessment' and 'Assignment'. The following Table 6.2 shows the independent variables, Assessment and Assignment, and the target variable Exam Result with their values. Draw a binary decision tree.

Table 6.2: Attributes and Associated Values

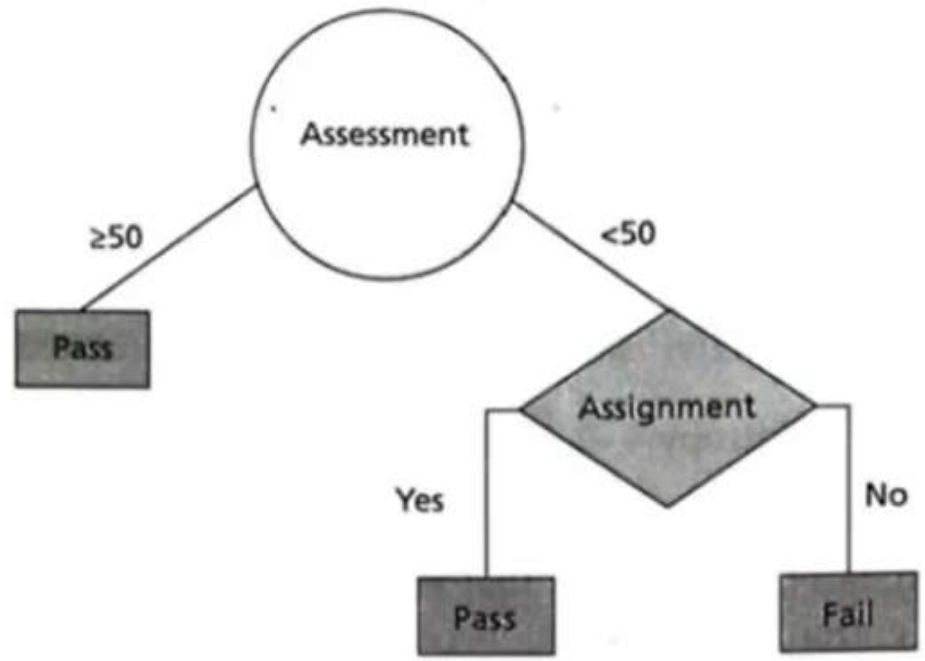
Attributes	Values
Assessment	≥50, <50
Assignment	Yes, No
Exam Result	Pass, Fail

- Consider root node as assessment
- If marks >= 50, root node is branched to leaf node Pass, else, branch to another decision node
- If students has submitted assessment, node branches to Pass , else fail

This tree can be interpreted as a sequence of logical rules as follows:

```
if (Assessment ≥ 50) then 'Pass'
else if (Assessment < 50) then
    if (Assignment == Yes) then 'Pass'
    else if (Assignment == No) then 'Fail'
```

- ***If a test instance is given, such as a student has scored 42 marks, and has not submitted assessment, then his result if 'Fail'***



Fundamentals of Entropy

- Entropy is the amount of **uncertainty or randomness** in the outcome of random variable or an event
- It describes how homogeneity of the data instance
- Best feature is selected based on the entropy value

What is the formula for entropy?

Entropy (H) is calculated using the formula:

$$H = -\sum(p_i \log_2 p_i)$$

Where p_i is the probability of each outcome occurring.

Fundamentals of Entropy

- *Eg: a coin is flipped , head or tail are 2 outcomes. Hence entropy of lower when compared to rolling a dice which has 6outcomes*
- Eg1: if all instances are **homogeneous say (1,0)**
  - It means all instances belong to same class , then **entropy is 0**
- Eg2: If instances are equally distributed, say (0.5,0.5), i.e., 50% positive , 50% negative, then **entropy is 1**

Eg3: If there are 10 data instances, where 6 belongs to class 1 and 4 belongs to class 0, then entropy is

$$\text{Entropy} = -\left[\frac{6}{10}\log_2\frac{6}{10} + \frac{4}{10}\log_2\frac{4}{10}\right]$$

Key points on Entropy

- If the dataset with complete homogeneity has an entropy of zero
- Elseif the dataset samples are equally divided(say 50-50) has an entropy of one
- Else, the entropy lies between 0 &1

Eg3: If there are 10 data instances, where 6 belongs to class 1 and 4 belongs to class 0, then entropy is

$$\text{Entropy} = -\left[\frac{6}{10}\log_2\frac{6}{10} + \frac{4}{10}\log_2\frac{4}{10}\right]$$

Information Gain

- Information gain is the mathematical difference in entropy before and after you split the data by a specific attribute.
- The information gain is based on the **decrease in entropy after a data-set is split on an attribute**.
- Constructing a decision tree is about finding the attribute that returns the highest information gain (i.e., the most homogeneous branches)

Information gain is the mathematical difference in entropy before and after you split the data by a specific attribute.

$$\text{Information Gain} = \text{Entropy(Before Split)} - \text{Entropy(After Split)}$$

- An attribute with **low information gain** leaves the resulting branches highly mixed and chaotic.
- An attribute with **high information gain** significantly drops the uncertainty, creating highly organized, predictable, and homogeneous branches.

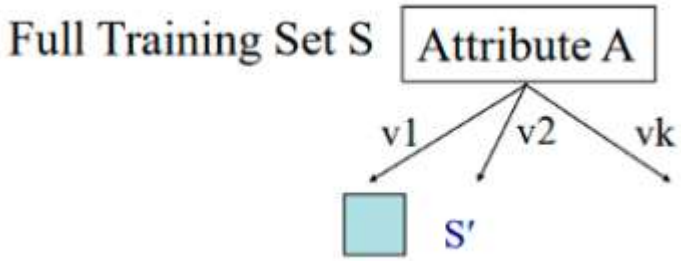
When a branch **is perfectly homogeneous, all the data points in that branch belong to the same category or class (e.g., a group containing only "Yes" answers rather than** a mix of "Yes" and "No").

Information Gain

The information gain is based on the decrease in entropy after a data-set is split on an attribute.

Constructing a decision tree is about finding attribute that returns the highest information gain (i.e., the most homogeneous branches)

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in D_A} \frac{|S_v|}{|S|} Entropy(S_v)$$



Assume that using attribute A, the set S will be partitioned into sets S1, S2 , …, Sv
v is number of values of the attribute A

S′={s∈S | value(A)=v1}

| Sv | = sum of every node values.
Sv is the v-th sub-dataset after being split.

| S | = sum of set S node values
S is the original dataset

Entropy(Sv) = entropy of current node. The Entropy of the v-th sub-dataset

| S | and | Sv | are the numbers of samples belong to the original dataset and the sub-dataset, respectively

ID3 Algorithms Intuition

- The ID3 algorithm builds decision trees using a top-down greedy search approach through the space of possible branches with no backtracking.
- A greedy algorithm, as the name suggests, always makes the choice that seems to be the best at that moment

Steps in ID3 algorithm

1. Select Root node(S) based on low Entropy and Highest Information Gain

2. On each iteration, calculate the Entropy and Information gain (IG), considering that every node is unused

3. Select the node based on the lowest entropy or Highest IG

4. The set S is then split by the selected attribute to produce a subset of the data

5. The algorithm continues to recur on each subset, considering only attributes never selected before

ID3 Problem Statement

Example 6.3: Assess a student's performance during his course of study and predict wheth a student will get a job offer or not in his final year of the course. The training dataset *T* consis of 10 data instances with attributes such as 'CGPA', 'Interactiveness', 'Practical Knowledge' ar 'Communication Skills' as shown in Table 6.3. The target class attribute is the 'Job Offer'.

- Build a decision tree using ID3 algorithm for the given training and predict the class of the following new example:
- Based on the attributes given, we have to find whether the student gets a job offer or not

No	CGPA	INTER ACTIVE NESS	PRACTICAL KNOWLEDGE	COMMUNICATI ON SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

ID3 Problem Statement

Step 1: Calculate entropy for target class **'Job Offer'**

Entropy_Info(Target Attribute = Job Offer) = Entropy_Info(7, 3) =

$$= -\left[\frac{7}{10}\log_2\frac{7}{10} + \frac{3}{10}\log_2\frac{3}{10}\right] = -(-0.3599 + -0.5208) = 0.8807$$

No	CGPA	INTER ACTIVE NESS	PRACTICAL KNOWLEDGE	COMMUNICATI ON SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

ID3 Problem Statement

No	CGPA	INTERACTIVENESS	PRACTICAL KNOWLEDGE	COMMUNICATION SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

Iteration 1:

Step 2: Calculate **entropy_info** and **information gain** for each attribute

Attribute: CGPA

CGPA	Job Offer = Yes	Job Offer = No	Total	Entropy
≥9	3	1	4	
≥8	4	0	4	0
<8	0	2	2	0

Entropy_Info(T, CGPA)

$$= \frac{4}{10}\left[-\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4}\right] + \frac{4}{10}\left[-\frac{4}{4}\log_2\frac{4}{4} - \frac{0}{4}\log_2\frac{0}{4}\right] + \frac{2}{10}\left[-\frac{0}{2}\log_2\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2}\right]$$

$$= \frac{4}{10}(0.3111 + 0.4997) + 0 + 0$$

$$= 0.3243$$

- Information Gain for CGPA=0.8807-0.3243
  = 0.5564

ID3 Problem Statement

- Repeat computing the entropy value and IG for the remaining attributes:

Attribute: Interactiveness

Table 6.5: Entropy Information for Interactiveness

Interactiveness	Job Offer = Yes	Job Offer = No	Total	Entropy
YES	5	1	6	
NO	1	2	4	

$$\text{Entropy_Info}(T, \text{Interactiveness}) = \frac{6}{10}\left[-\frac{5}{6}\log_2\frac{5}{6} - \frac{1}{6}\log_2\frac{1}{6}\right] + \frac{4}{10}\left[-\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}\right]$$

$$= \frac{6}{10}(0.2191 + 0.4306) + \frac{4}{10}(0.4997 + 0.4997)$$

$$= 0.3898 + 0.3998 = 0.7896$$

$$\text{Gain(Interactiveness)} = 0.8807 - 0.7896$$

$$= 0.0911$$

No	CGPA	INTERACTIVENESS	PRACTICAL KNOWLEDGE	COMMUNICATION SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

ID3 Problem Statement

No	CGPA	INTERACTIVENESS	PRACTICAL KNOWLEDGE	COMMUNICATION SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

Attribute: Practical knowledge

Table 6.6: Entropy Information for Practical Knowledge

Practical Knowledge	Job Offer = Yes	Job Offer = No	Total	Entropy
Very Good	2	0	2	0
Average	1	2	3	
Good	4	1	5	

Entropy_Info(T, Practical Knowledge)

$$= \frac{2}{10}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{3}{10}\left[-\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}\right] + \frac{5}{10}\left[-\frac{4}{5}\log_2\frac{4}{5} - \frac{1}{5}\log_2\frac{1}{5}\right.$$

$$= \frac{2}{10}(0) + \frac{3}{10}(0.5280 + 0.3897) + \frac{5}{10}(0.2574 + 0.4641)$$

$$= 0 + 0.2753 + 0.3608$$

$$= 0.6361$$

$$\text{Gain(Practical Knowledge)} = 0.8807 - 0.6361$$

$$= 0.2446$$

ID3 Problem Statement

Attribute: Communication skills

Table 6.7: Entropy Information for Communication Skills

Communication Skills	Job Offer = Yes	Job Offer = No	Total
Good	4	1	5
Moderate	3	0	3
Poor	0	2	2

Entropy_Info(T, Communication Skills)

$$= \frac{5}{10}\left[-\frac{4}{5}\log_2\frac{4}{5} - \frac{1}{5}\log_2\frac{1}{5}\right] + \frac{3}{10}\left[-\frac{3}{3}\log_2\frac{3}{3} - \frac{0}{3}\log_2\frac{0}{3}\right] + \frac{2}{10}\left[-\frac{0}{2}\log_2\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2}\right]$$

$$= \frac{5}{10}(0.5280 + 0.3897) + \frac{3}{10}(0) + \frac{2}{10}(0)$$

$$= 0.3609$$

$$\text{Gain(Communication Skills)} = 0.8813 - 0.36096$$

$$= 0.5203$$

No	CGPA	INTER ACTIVE NESS	PRACTICAL KNOWLEDGE	COMMUNICATI ON SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

Now summarize the Information gain of all attributes:

Table 6.8: Gain

Attributes	Gain
CGPA	0.5564
Interactiveness	0.0911
Practical Knowledge	0.2246
Communication Skills	0.5203

he Gain calculated for all the attributes is shown in Table 6.8:

Table 6.8: Gain

Attributes	Gain
CGPA	**0.5564**
Interactiveness	0.0911
Practical Knowledge	0.2246
Communication Skills	0.5203

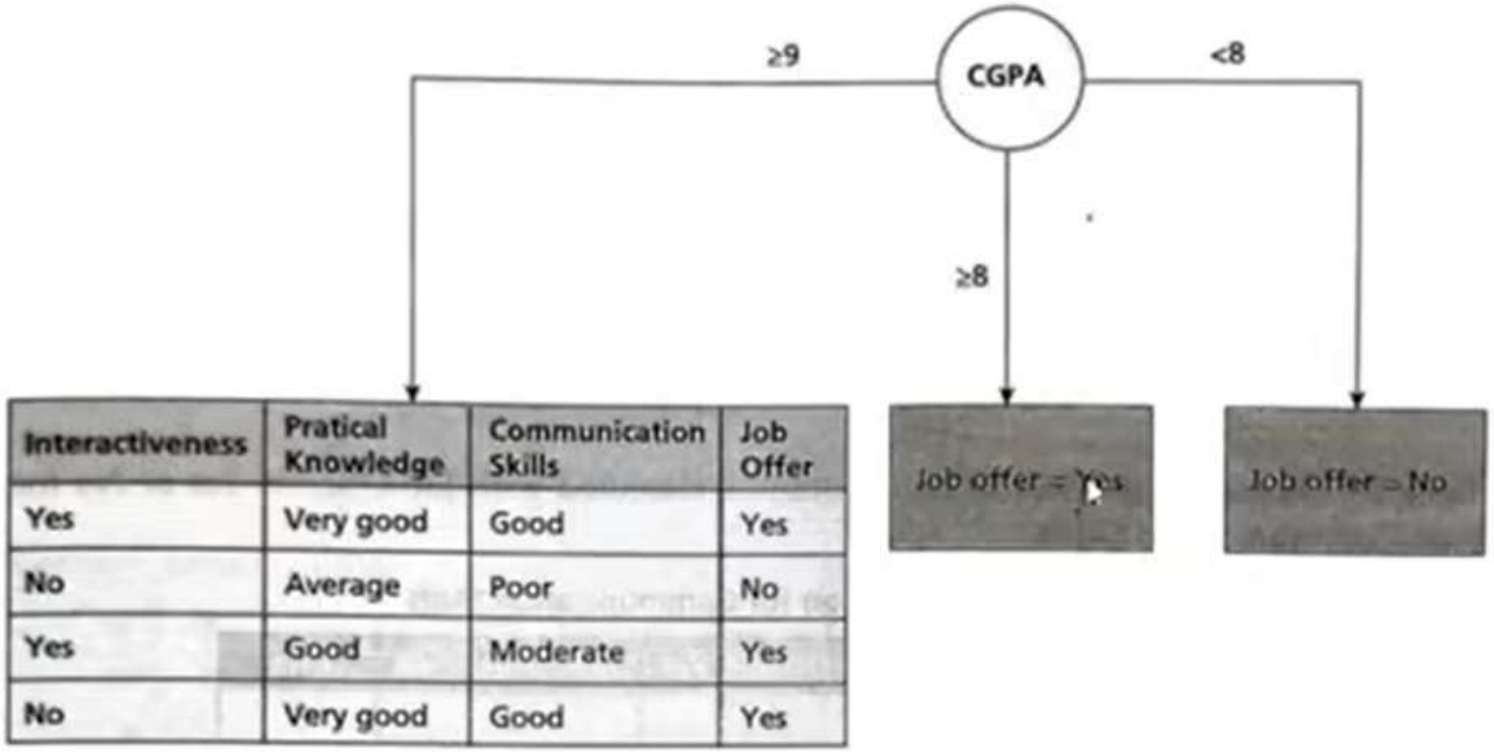
- **Step 3: Choose the attribute whose entropy is minimum and gain is maximum**. From Table 6.8., choose CGPA has the root node since it has maximum gain

  - When we choose CGPA as root node, the path can be split into 3: ≥ 9 ≥ 8 < 8

  - **For the path >=8** , all path instances are classified as JobOffer=YES

  - **For the path <8** , all path instances are classified as JobOffer=NO

- Whenever the CGPA is >= 8, the student always gets a job offer; otherwise, no job offer. These two CGPA values will end up as leaf nodes. So we don't have to branch further
- But whenever the CGPA is >= 9, the student may or may not get a job offer, so further splitting is required (sometimes "yes" , sometimes "n").
- So for >=9, take all remaining attributes. Based upon the new table, further splitting is required



Interactiveness	Pratical Knowledge	Communication Skills	Job Offer
Yes	Very good	Good	Yes
No	Average	Poor	No
Yes	Good	Moderate	Yes
No	Very good	Good	Yes

Figure 6.3: Decision Tree After Iteration 1

- So for >=9, take all remaining attributes. Based upon the new table, further splitting is required
- Find the total entropy for the new table

- ***Now, continue the process for subset of data instances of CGPA>=9***
- **Iteration 2:**
- **The same** process of computing entropy_info and information_gain is repeated for the

Subset of training set

- The subset of 4 data instances is shown:

Interactiveness	Pratical Knowledge	Communication Skills	Job Offer
Yes	Very good	Good	Yes
No	Average	Poor	No
Yes	Good	Moderate	Yes
No	Very good	Good	Yes

- Attribute: interactiveness

Entropy information for interactiveness

Interactiveness	Job offer=yes	Job offer=no	Total
Yes	2	0	2
no	1	1	2

$$\text{Entropy_Info}(T) = \text{Entropy_Info}(3, 1) =$$

$$= -\left[\frac{3}{4}\log_2\frac{3}{4} + \frac{1}{4}\log_2\frac{1}{4}\right]$$

$$= -(-0.3111 + -0.4997)$$

$$= 0.8108$$

- So for >=9, take all remaining attributes. Based upon the new table, further splitting is required
- Find the total entropy for the new table

$$\text{Entropy_Info}(T, \text{Interactiveness}) = \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{2}{4}\left[-\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}\right]$$

$$= 0 + 0.4997$$

$$\text{Gain(Interactiveness)} = 0.8108 - 0.4997$$

=3.111

- Next repeat the entropy and Gain calculation for practical knowledge and communication skills
-

- **Attribute: practical knowledge**

Entropy information for attr: practical knowledge

PRACTICAL KNOWLEDGE	Job offer=yes	Job offer=no	Total
VERY GOOD	2	0	2
AVERAGE	0	1	1
GOOD	1	0	1

$$\text{Entropy_Info}(T, \text{Practical Knowledge}) = \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{1}{4}\left[-\frac{0}{1}\log_2\frac{0}{1} - \frac{1}{1}\log_2\frac{1}{1}\right] + \frac{1}{4}\left[-\frac{1}{1}\log_2\frac{1}{1} - \frac{0}{1}\log_2\frac{0}{1}\right]$$

$$= 0$$

$$\text{Gain(Practical Knowledge)} = 0.8108$$

- Next repeat the entropy and Gain calculation for communication skills
-

- **Attribute: communication skills**

COMMUNICATION SKILLS	Job offer=yes	Job offer=no	Total
GOOD	2	0	2
POOR	0	1	1
MODERATE	1	0	1

$$\text{Entropy_Info}(T, \text{Communication Skills}) = \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{1}{4}\left[-\frac{0}{1}\log_2\frac{0}{1} - \frac{1}{1}\log_2\frac{1}{1}\right] + \frac{1}{4}\left[-\frac{1}{1}\log_2\frac{1}{1} - \frac{0}{1}\log_2\frac{0}{1}\right]$$

$$= 0$$

$$\text{Gain(Communication Skills)} = 0.8108$$

- Summary of gain information:
- Both communication skills and practical knowledge have the same gain, so we can choose either of them

- **Gain for all attributes:**

ATTRIBUTES	GAIN
Interactiveness	0.3111
PRACTICAL KNOWLEDGE	0.8108
COMMUNICATION SKILLS	0.8108

Entropy information for attr: practical knowledge

PRACTICAL KNOWLEDGE	Job offer=yes	Job offer=no	Total
VERY GOOD	2	0	2
AVERAGE	0	1	1
GOOD	1	0	1

- Construct the final decision tree
- Based on Practical knowledge (selected attribute), the final decision tree is shown below
- Whenever the practical knowledge is very good & good job offer = yes; otherwise, job offer=no

Gain for all attributes:

ATTRIBUTES	GAIN
Interactiveness	0.3111
PRACTICAL KNOWLEDGE	0.8108
COMMUNICATION SKILLS	0.8108

- Both PRACTICAL KNOWLEDGE and COMMUNICATION SKILLS have same higher gain

- Either construct decision tree using PRACTICAL KNOWLEDGE or COMMUNICATION SKILLS

- The decision tree when split is based on PRACTICAL KNOWLEDGE

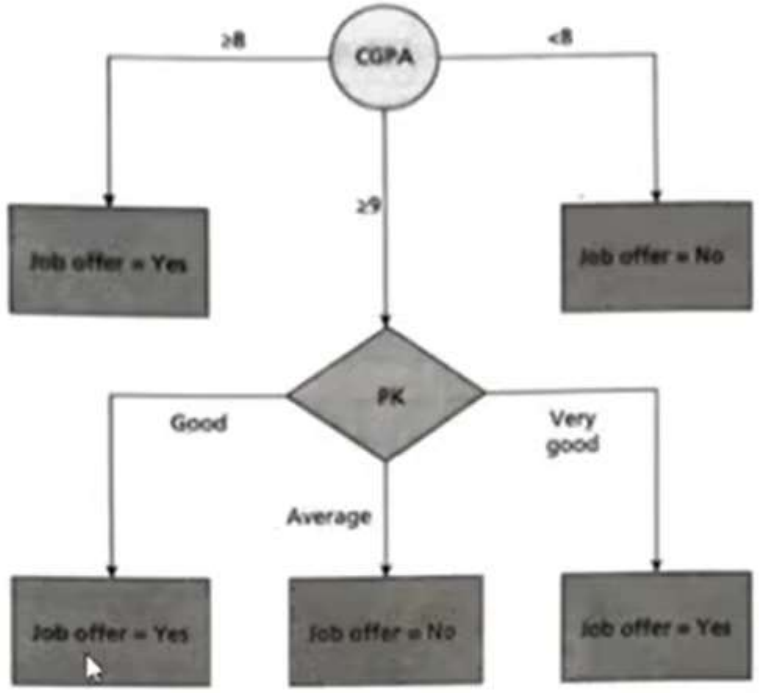


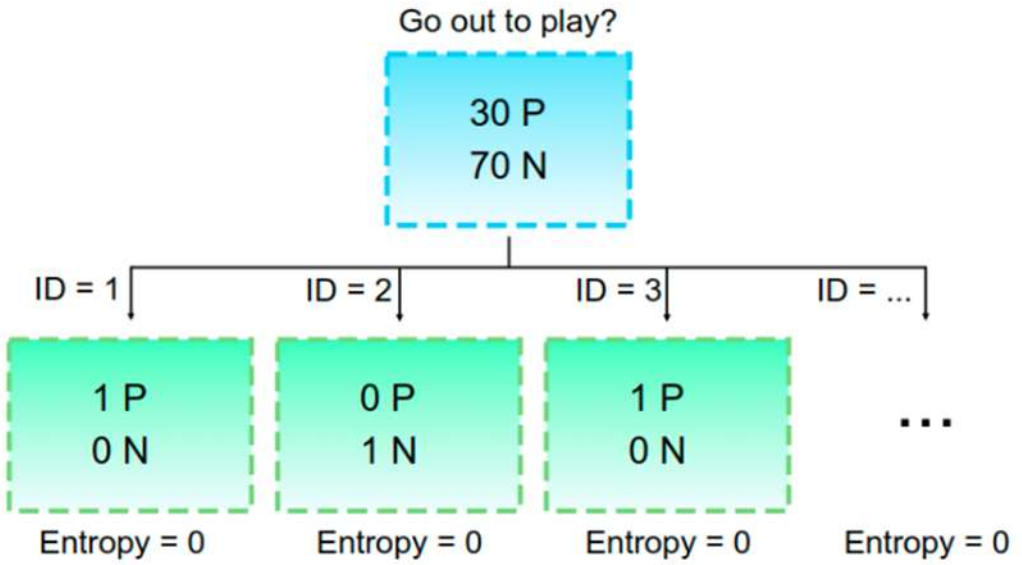
Figure 6.6: Final Decision Tree

Decision Tree using C4.5 Algorithm

- C4.5, an improvement of ID3
- C4.5 uses the Gain Ratio to split the dataset, unlike ID3, which used the Information Gain.
- **Gain Ratio** corrects this bias by introducing a penalty called **Split Info** (Split Entropy). Split Info measures how much the attribute itself splits the data.
- Gain ratio overcomes the problem with information gain by taking into account the number of branches that would result from the split.
- It corrects information gain by taking the split information or intrinsic information of a split into account.

Information Gain – Undesirable Characteristic

- The Information Gain has an undesired characteristic
- It favors the predictor variables with a large number of values.
- It means it prefers the attribute with a large number of distinct values
- Highly branching predictors are likely to split the data into subsets with low Entropy values.
- Ex1: in the extreme case, the ID code



Information Gain – Undesirable Characteristic

- Ex2: Suppose we have two features, “Color” and “Size”
- To build a decision tree to predict the type of fruit based on these two features.
- The “Color” feature has three outcomes (red, green, yellow) and the “Size” feature has two outcomes (small, large).
- Using the information gain method, the “Color” feature would be chosen as the best feature to split on because it has the highest information gain.
- However, this could be a problem because the “Size” feature could be a better feature to split on because it is less ambiguous and has fewer outcomes.

Information Gain – Undesirable Characteristic

- The disadvantages of these splits are:
  - Making the model more prone to over-fitting (selection of an attribute that is non-optimal for prediction)
  - The number of nodes in the tree may be very large.
- To address this issue, an adjusted version of Information Gain called Gain Ratio is used

C4.5 Decision Tree

- C4.5 is an improvement over ID3 algorithm
- Works for continuous and discrete attributes and missing values
- It supports post-pruning
- It uses gain-ratio for selecting the best-split attribute

C4.5 Decision Tree

Gain Ratio:

The Gain_Ratio of an attribute A is computed as

$$\text{Gain_Ratio}(A) = \frac{\text{Info_Gain}(A)}{\text{Split_Info}(T, A)}$$

The Split_Info of an attribute A is computed as

$$\text{Split_Info}(T, A) = -\sum_{i=1}^{v} \frac{|A_i|}{|T|} \times \log_2 \frac{|A_i|}{|T|}$$

where, the attribute A has got 'v' distinct values $\{a_1, a_2, \ldots. a_v\}$, and $|A_i|$ is the number of instanc for distinct value 'i' in attribute A.

Steps in C4.5 algorithm

1. Calculate the Entropy_information for the entire training dataset using:

$$\text{Entropy_Info}(T) = -\sum_{i=1}^{m} P_i \log_2 P_i$$

2. Compute the entropy_info, information gain, and split_info and gain_ratio for each attribute in the training dataset

4. Choose the attribute for which gain_ratio is maximum as the best split attribute

5. The best split attribute is chosen as the root node

6. Next, the root node is split into subtrees as an outcome of test condition

7. Recursively apply the same operation for the subset of training dataset until leaf node is derived and no more training instances are available

C4.5 Problem Statement

Example 6.3: Assess a student's performance during his course of study and predict wheth a student will get a job offer or not in his final year of the course. The training dataset *T* consis of 10 data instances with attributes such as 'CGPA', 'Interactiveness', 'Practical Knowledge' ar 'Communication Skills' as shown in Table 6.3. The target class attribute is the 'Job Offer'.

- Build a decision tree using C4.5 algorithm for the given training and predict the class of the following new example:
- Based on the attributes given, we have to find whether the student gets a job offer or not
- The first step is same as ID3

No	CGPA	INTER ACTIVE NESS	PRACTICAL KNOWLEDGE	COMMUNICATI ON SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

C4.5 Problem Statement

Step 1: Calculate entropy for target class '**Job Offer**'

Entropy_Info(Target Attribute = Job Offer) = Entropy_Info(7, 3) =

$$= -\left[\frac{7}{10}\log_2\frac{7}{10} + \frac{3}{10}\log_2\frac{3}{10}\right] = -(-0.3599 + -0.5208) = 0.8807$$

No	CGPA	INTERACTIVENESS	PRACTICAL KNOWLEDGE	COMMUNICATION SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

C4.5 Problem Statement

Step 1: Calculate entropy for target class **'Job Offer'**

Entropy_Info(Target Attribute = Job Offer) = Entropy_Info(7, 3) =

$$= -\left[\frac{7}{10}\log_2\frac{7}{10} + \frac{3}{10}\log_2\frac{3}{10}\right] = -(-0.3599 + -0.5208) = 0.8807$$

No	CGPA	INTER ACTIVE NESS	PRACTICAL KNOWLEDGE	COMMUNICATION SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

C4.5 Problem Statement

Step 2: Calculate the Entropy_Info, Gain(Info_Gain), Split_Info, Gain_Ratio for each of the attribute in the training dataset.

No	CGPA	INTERACTIVENESS	PRACTICAL KNOWLEDGE	COMMUNICATION SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

C4.5 Problem Statement

Step 2: Calculate the Entropy_Info, Gain(Info_Gain), Split_Info, Gain_Ratio for each of the attribute in the training dataset.

No	CGPA	INTERACTIVENESS	PRACTICAL KNOWLEDGE	COMMUNICATION SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

C4.5 Problem Statement

No	CGPA	INTER ACTIVE NESS	PRACTICAL KNOWLEDGE	COMMUNICATION SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

Iteration 1:

Step 2: Calculate **entropy_info** and **information gain** for each attribute

Attribute: CGPA

CGPA	Job Offer = Yes	Job Offer = No	Total	Entropy
≥9	3	1	4	
≥8	4	0	4	0
<8	0	2	2	0

Entropy_Info(T, CGPA)

$$= \frac{4}{10}\left[-\frac{3}{4}\log_2\frac{3}{4}-\frac{1}{4}\log_2\frac{1}{4}\right]+\frac{4}{10}\left[-\frac{4}{4}\log_2\frac{4}{4}-\frac{0}{4}\log_2\frac{0}{4}\right]+\frac{2}{10}\left[-\frac{0}{2}\log_2\frac{0}{2}-\frac{2}{2}\log_2\frac{2}{2}\right]$$

$$= \frac{4}{10}(0.3111+0.4997)+0+0$$

$$= 0.3243$$

Information Gain for CGPA=0.8807-0.3243=0.5564

$$\text{Split_Info}(T, \text{CGPA}) = -\frac{4}{10}\log_2\frac{4}{10}-\frac{4}{10}\log_2\frac{4}{10}-\frac{2}{10}\log_2\frac{2}{10}$$
$$= 0.5285 + 0.5285 + 0.4641$$
$$= 1.5211$$

Gain Ratio=0.5564/1.5211=0.3658

$$\text{Gain Ratio(CGPA)} = (\text{Gain(CGPA)})/(\text{Split_Info}(T, \text{CGPA}))$$

C4.5 Problem Statement

- Repeat computing the entropy value IG , split_info and Gain_ratiofor the remaining attributes:

No	CGPA	INTER ACTIVE NESS	PRACTICAL KNOWLEDGE	COMMUNICATI ON SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

Attribute: Interactiveness

Table 6.5: Entropy Information for Interactiveness

Interactiveness	Job Offer = Yes	Job Offer = No	Total	Entropy
YES	5	1	6	
NO	1	2	4	

$$\text{Entropy_Info}(T, \text{Interactiveness}) = \frac{6}{10}\left[-\frac{5}{6}\log_2\frac{5}{6} - \frac{1}{6}\log_2\frac{1}{6}\right] + \frac{4}{10}\left[-\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}\right]$$

$$= \frac{6}{10}(0.2191 + 0.4306) + \frac{4}{10}(0.4997 + 0.4997)$$

$$= 0.3898 + 0.3998 = 0.7896$$

$$\text{Gain(Interactiveness)} = 0.8807 - 0.7896$$

$$= 0.0911$$

$$split_info(T, interactiveness) = \frac{-6}{10}\log(\frac{6}{10})\frac{-4}{10}\log(\frac{4}{10}) = 0.9704$$

Gain Ratio=0.0911/0.9704=0.0939

C4.5 Problem Statement

No	CGPA	INTERACTIVENESS	PRACTICAL KNOWLEDGE	COMMUNICATION SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

Attribute: Practical knowledge

Table 6.6: Entropy Information for Practical Knowledge

Practical Knowledge	Job Offer = Yes	Job Offer = No	Total	Entropy
Very Good	2	0	2	0
Average	1	2	3	
Good	4	1	5	

$$\text{Entropy_Info}(T, \text{Practical Knowledge}) = \frac{2}{10}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{3}{10}\left[-\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}\right] + \frac{5}{10}\left[-\frac{4}{5}\log_2\frac{4}{5} - \frac{1}{5}\log_2\frac{1}{5}\right]$$

$$= \frac{2}{10}(0) + \frac{3}{10}(0.5280 + 0.3897) + \frac{5}{10}(0.2574 + 0.4641)$$

$$= 0 + 0.2753 + 0.3608$$

$$= 0.6361$$

$$\text{Gain(Practical Knowledge)} = 0.8807 - 0.6361 = 0.2446$$

$$\text{Split_Info}(T, \text{Practical Knowledge}) = -\frac{2}{10}\log_2\frac{2}{10} - \frac{5}{10}\log_2\frac{5}{10} - \frac{3}{10}\log_2\frac{3}{10} = 1.4853$$

Gain Ratio=0.2446/1.4853=0.1648

C4.5 Problem Statement

Attribute: Communication skills

Table 6.7: Entropy Information for Communication Skills

Communication Skills	Job Offer = Yes	Job Offer = No	Total
Good	4	1	5
Moderate	3	0	3
Poor	0	2	2

$$\text{Entropy_Info}(T, \text{Communication Skills})$$

$$= \frac{5}{10}\left[-\frac{4}{5}\log_2\frac{4}{5} - \frac{1}{5}\log_2\frac{1}{5}\right] + \frac{3}{10}\left[-\frac{3}{3}\log_2\frac{3}{3} - \frac{0}{3}\log_2\frac{0}{3}\right] + \frac{2}{10}\left[-\frac{0}{2}\log_2\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2}\right]$$

$$= \frac{5}{10}(0.5280 + 0.3897) + \frac{3}{10}(0) + \frac{2}{10}(0)$$

$$= 0.3609$$

$$\text{Gain(Communication Skills)} = 0.8813 - 0.36096$$

$$= 0.5203$$

No	CGPA	INTERACTIVENESS	PRACTICAL KNOWLEDGE	COMMUNICATION SKILLS	JOB OFFER
1	≥ 9	YES	VERY GOOD	GOOD	YES
2	≥ 8	NO	GOOD	MODERATE	YES
3	≥ 9	NO	AVERAGE	POOR	NO
4	< 8	NO	AVERAGE	GOOD	NO
5	≥ 8	YES	GOOD	MODERATE	YES
6	≥ 9	YES	GOOD	MODERATE	YES
7	< 8	YES	GOOD	POOR	NO
8	≥ 9	NO	VERY GOOD	GOOD	YES
9	≥ 8	YES	GOOD	GOOD	YES
10	≥ 8	YES	AVERAGE	GOOD	YES

$$\text{Split_Info}(T, \text{Communication Skills}) = -\frac{5}{10}\log_2\frac{5}{10} - \frac{3}{10}\log_2\frac{3}{10} - \frac{2}{10}\log_2\frac{2}{10}$$

$$= 1.4853$$

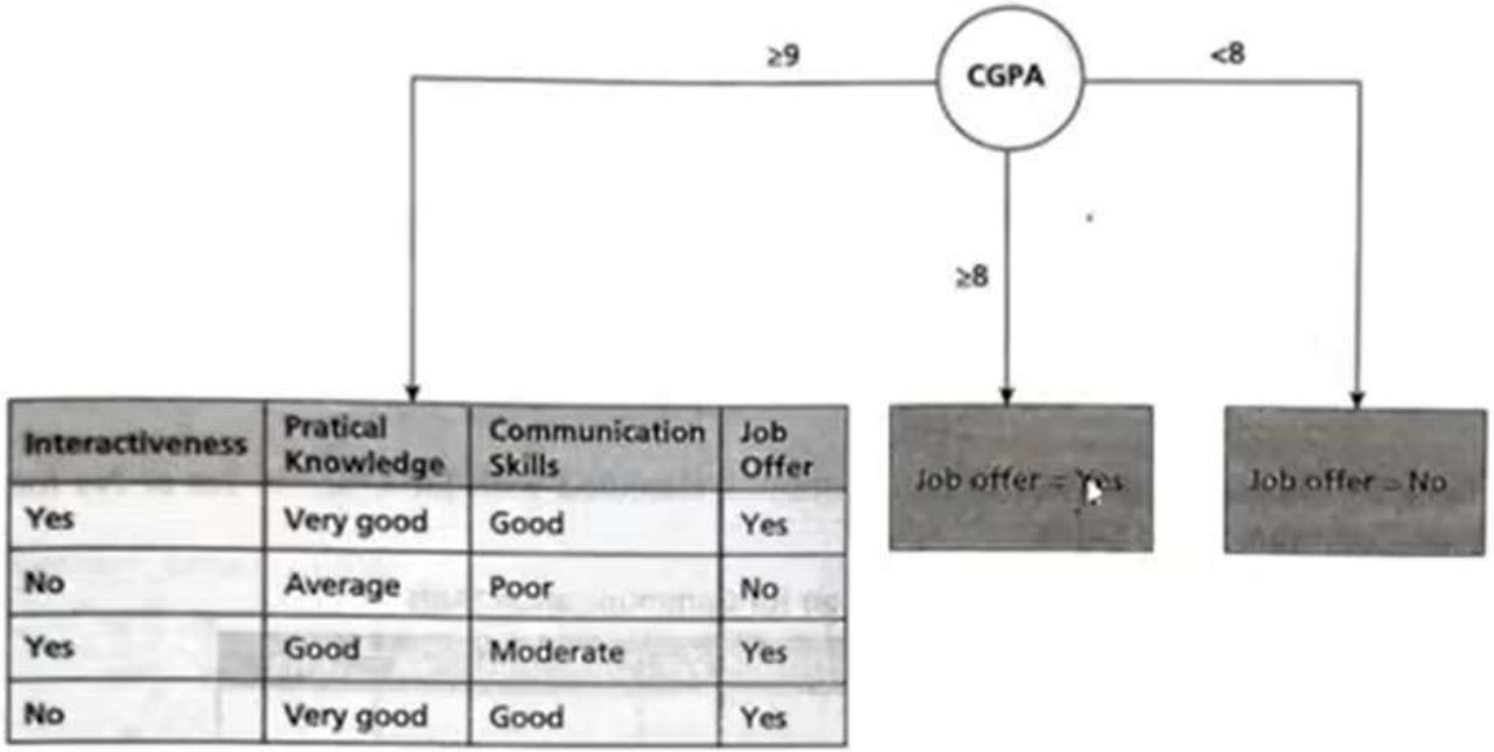
Gain Ratio=0.5208/1.4853=0.3502

- Now summarize the **gain ratio** of all attributes:

Attribute	Gain_Ratio
CGPA	**0.3658**
INTERACTIVENESS	0.0939
PRACTICAL KNOWLEDGE	0.1648
COMMUNICATION SKILLS	0.3502

- Choose the one with the maximum gain ratio as the best split attribute.
- So CGPA is chosen as root node and branching is done

- Whenever the CGPA is >= 8, the student always gets a job offer; otherwise, no job offer. These two CGPA values will end up as leaf nodes. So we don't have to branch further
- But whenever the CGPA is >= 9, the student may or may not get a job offer, so further splitting is required (sometimes "yes" , sometimes "n").
- So for >=9, take all remaining attributes. Based upon the new table, further splitting is required



Interactiveness	Pratical Knowledge	Communication Skills	Job Offer
Yes	Very good	Good	Yes
No	Average	Poor	No
Yes	Good	Moderate	Yes
No	Very good	Good	Yes

Figure 6.3: Decision Tree After Iteration 1

- So, for >= 9, take all remaining attributes. Based upon the new table, further splitting is required to choose the subnode out of 3 attributes (Practical knowledge, communication skill &Interactiveness)
- Find the total entropy for the new table (with 3 attributes)

Iteration 2:

Repeat the same process for this resultant dataset with 4 data instances.

Job Offer has 3 instances as Yes and 1 instance as No.

Interactiveness	Pratical Knowledge	Communication Skills	Job Offer
Yes	Very good	Good	Yes
No	Average	Poor	No
Yes	Good	Moderate	Yes
No	Very good	Good	Yes

$$\text{Entropy_Info(Target Class = Job Offer)} = -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4}$$
$$= 0.3112 + 0.5$$
$$= 0.8112$$

- Find the entropy for the first attribute: Interactiveness

Interactiveness	Job offer=yes	Job offer=no	Total
Yes	2	0	2
no	1	1	2

$$\text{Entropy_Info}(T, \text{Interactiveness}) = \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{2}{4}\left[-\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}\right]$$

$$= 0 + 0.4997$$

$$\text{Gain(Interactiveness)} = 0.8108 - 0.4997$$

=3.112

$$\text{Split_Info}(T, \text{Interactiveness}) = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4} = 0.5 + 0.5 = 1$$

Gain Ratio=3.112/1=0.3112

- Next repeat the entropy and Gain calculation for practical knowledge and communication skills
-

- **Attribute: practical knowledge**

Entropy information for attr: practical knowledge

PRACTICAL KNOWLEDGE	Job offer=yes	Job offer=no	Total
VERY GOOD	2	0	2
AVERAGE	0	1	1
GOOD	1	0	1

$$\text{Entropy_Info}(T, \text{Practical Knowledge}) = \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{1}{4}\left[-\frac{0}{1}\log_2\frac{0}{1} - \frac{1}{1}\log_2\frac{1}{1}\right] + \frac{1}{4}\left[-\frac{1}{1}\log_2\frac{1}{1} - \frac{0}{1}\log_2\frac{0}{1}\right]$$

$$= 0$$

$$\text{Gain(Practical Knowledge)} = 0.8108$$

$$\text{Split_Info}(T, \text{Practical Knowledge}) = -\frac{2}{4}\log_2\frac{2}{4} - \frac{1}{4}\log_2\frac{1}{4} - \frac{1}{4}\log_2\frac{1}{4} = 1.5$$

Gain Ratio=0.8108/1.5=0.5408

- Next repeat the entropy and Gain calculation for communication skills
-

- **Attribute: communication skills**

COMMUNICATION SKILLS	Job offer=yes	Job offer=no	Total
GOOD	2	0	2
POOR	0	1	1
MODERATE	1	0	1

$$\text{Entropy_Info}(T, \text{Communication Skills}) = \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{1}{4}\left[-\frac{0}{1}\log_2\frac{0}{1} - \frac{1}{1}\log_2\frac{1}{1}\right] + \frac{1}{4}\left[-\frac{1}{1}\log_2\frac{1}{1} - \frac{0}{1}\log_2\frac{0}{1}\right]$$

$$= 0$$

$$\text{Gain(Communication Skills)} = 0.8108$$

$$\text{Split_Info}(T, \text{Communication Skills}) = -\frac{2}{4}\log_2\frac{2}{4} - \frac{1}{4}\log_2\frac{1}{4} - \frac{1}{4}\log_2\frac{1}{4} = 1.5$$

Gain Ratio=0.8108/1.5=0.5408

- Summary of gain ratio information:

Summary of Gain_ratio for all attributes

Attributes	Gain_Ratio
Interactiveness	0.3112
Practical Knowledge	0.5408
Communication Skills	0.5408

- **Both 'practical knowledge' and 'Communication skills' have same higher gain ratio**
- **Can choose any of these as the node for best split**
- **If we split based on 'practical knowledge' , the final decision tree as shown**

- Construct the final decision tree
- Based on Practical knowledge (selected attribute), the final decision tree is shown below
- Whenever the practical knowledge is very good & good job offer = yes; otherwise, job offer=no

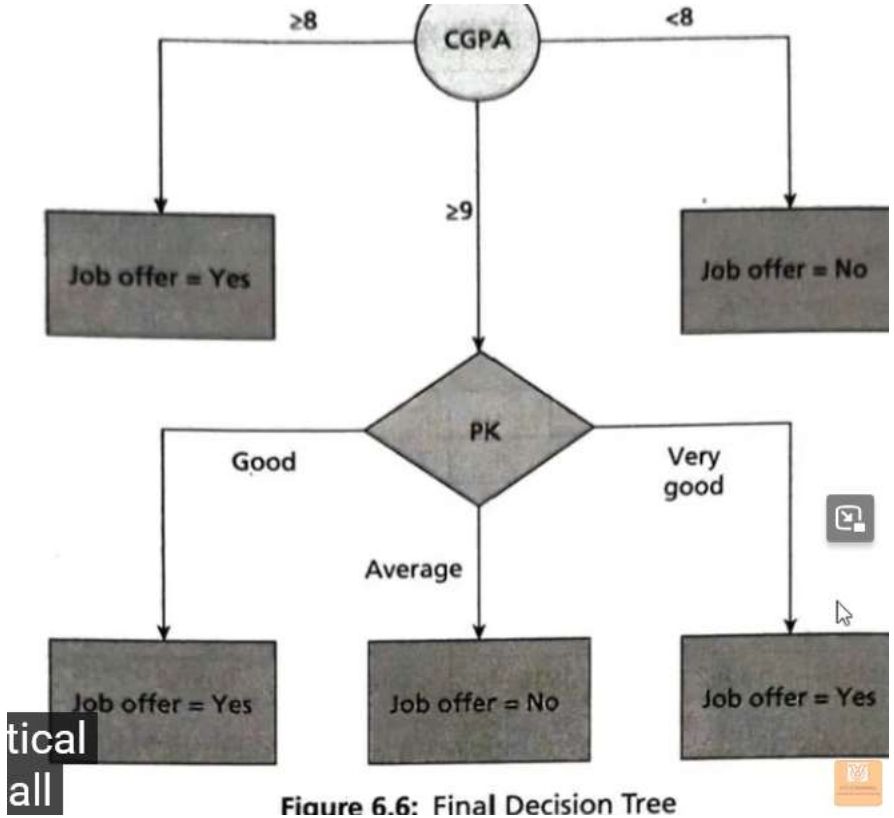


Figure 6.6: Final Decision Tree

- Now we can decide when to stop a decision tree as we don't need to have any more decision boxes

Procedure for constructing a decision tree using C4.5 for continuous values

Table 6.12: Sample Dataset

S.No.	CGPA	Job Offer
1.	9.5	Yes
2.	8.2	Yes
3.	9.1	No
4.	6.8	No
5.	8.5	Yes
6.	9.5	Yes
7.	7.9	No
8.	9.1	Yes
9.	8.8	Yes
10.	8.8	Yes

- C4.5 algorithm can be improved for continuous attributes
- Continuous attributes can be discretized by finding a split point or threshold
- For example in this case, attribute CGPA has continuous values

A decision tree using C4.5 for continuous values

- First we have to sort the values of CGPA in ascending order. Then remove the duplicates & consider only the unique values of the attributes
- Write down the range (<=, >) for all attributes (for example CGPA) then write the count: job offer= “yes” or “no” for each range of CGPA

	6.8		7.9		8.2		8.5		8.8		9.1		9.5	
Range	≤	>	≤	>	≤	>	≤	>	≤	>	≤	>	≤	>
Yes	0	7	0	7	1	6	2	5	4	3	5	2	7	0
No	1	2	2	1	2	1	2	1	2	1	3	0	3	0

Table 6.12: Sample Dataset

S.No.	CGPA	Job Offer
1.	9.5	Yes
2.	8.2	Yes
3.	9.1	No
4.	6.8	No
5.	8.5	Yes
6.	9.5	Yes
7.	7.9	No
8.	9.1	Yes
9.	8.8	Yes
10.	8.8	Yes

Procedure for constructing a decision tree using C4.5 for continuous values
Problem statement:

- First calculate entropy for the entire data set and then compute the entropy for each CGPA. Next compute Gain separately for each CGPA (for CGPA<=6.8 & CGPA>6.8 is shown below)

- **Calculate entropy, entropy_info(S,T) and gain for every distinct value of CGPA**
- **Lets start with CGPA : 6.8**

	6.8		7.9		8.2		8.5		8.8		9.1		9.5	
Range	≤	>	≤	>	≤	>	≤	>	≤	>	≤	>	≤	>
Yes	0	7	0	7	1	6	2	5	4	3	5	2	7	0
No	1	2	2	1	2	1	2	1	2	1	3	0	3	0
Entropy	0	0.7637												
Entropy_Info (S, T)	0.6873													
Gain	0.1935													

$$\text{Entropy_Info}(T, \text{Job_Offer}) = -\left[\frac{7}{10}\log_2\frac{7}{10} + \frac{3}{10}\log_2\frac{3}{10}\right]$$
$$= -(-0.3599 + -0.5209)$$
$$= 0.8808$$

$$\text{Entropy_Info}(T, \text{CGPA} \in 6.8) = \frac{1}{10} \times \text{Entropy}(0, 1) + \frac{9}{10}\text{Entropy}(7, 2)$$
$$= \frac{1}{10}\left[-\frac{0}{1}\log_2\frac{0}{1} - \frac{1}{1}\log_2\frac{1}{1}\right] + \frac{9}{10}\left[-\frac{7}{9}\log_2\frac{7}{9} - \frac{2}{9}\log_2\frac{2}{9}\right.$$
$$= 0 + \frac{9}{10}(0.7637)$$
$$= 0.6873$$

$$Entropy(0,1) = -\left[\frac{0}{1}\log\left(\frac{0}{1}\right) + \frac{1}{1}\log\left(\frac{1}{1}\right)\right] = 0$$

$$\text{Entropy}(7, 2) = -\left[\frac{7}{9}\log_2\frac{7}{9} + \frac{2}{9}\log_2\frac{2}{9}\right]$$
$$= -(-0.2818 + -0.4819)$$
$$= 0.7637$$

$$\text{Gain}(\text{CGPA} \in 6.8) = 0.8808 - 0.6873$$
$$= 0.1935$$

Procedure for constructing a decision tree using C4.5 for continuous values
Problem statement:

Repeat the entropy and gain calculation for each CGPA till 9.5

- **Calculate entropy, entropy_info(S,T) and gain for CGPA: 9.1**

	6.8		7.9		8.2		8.5		8.8		9.1		9.5	
Range	≤	>	≤	>	≤	>	≤	>	≤	>	≤	>	≤	>
Yes	0	7	0	7	1	6	2	5	4	3	5	2	7	0
No	1	2	2	1	2	1	2	1	2	1	3	0	3	0
Entropy	0	0.7637	0	0.5433	0.9177	0.5913	1	0.6497	0.9177	0.8108	0.9538	0		
Entropy_Info (S, T)	0.6873		0.4346		0.6892		0.7898		0.8749		0.7630			
Gain	0.1935		**0.4462**		0.1916		0.091		0.0059		0.1178			

$$entropy(5,3) = -\left[\frac{5}{8}log_2\left(\frac{5}{8}\right) + \frac{3}{8}log_2\left(\frac{3}{8}\right)\right] = 0.9538$$

$$entropy(2,0) = -\left[\frac{2}{2}log_2\left(\frac{2}{2}\right) + \frac{0}{2}log_2\left(\frac{0}{2}\right)\right] = 0$$

$$entropy_info(T, CGPA \in 9.1) = \frac{8}{10} * \text{entropy}(5,3) + \frac{2}{10} * \text{entropy}(2,0) = 0.7630$$

$$gain(CGPA \in 9.1) = 0.8808 - 0.7630 = 0.1178$$

$$\text{Entropy_Info}(T, \text{Job_Offer}) = -\left[\frac{7}{10}\log_2\frac{7}{10} + \frac{3}{10}\log_2 \right.$$

$$= -(-0.3599 + -0.5209)$$

$$= 0.8808$$

- After finding entropy and gain for ranges of CGPA, then summarize the gain

Summary of Gain_ratio for all attributes

	6.8	7.9	8.2	8.5	8.8	9.1	9.5
Gain	0.1935	**0.4462**	0.1916	0.091	0.0059	0.1178	0

- From the above table we can observe that CGPA 7.9 has the maximum gain. So we can choose CGPA=7.9 as the best split.
- Now we can discretize the continuous CGPA into two categories:

  CGPA<=7.9, CGPA>7.9

- The following table shows the discretized values of CGPA:

Table 6.14: Discretized Instances

S.No.	CGPA Continuous	CGPA Discretized	Job Offer
1.	9.5	>7.9	Yes
2.	8.2	>7.9	Yes
3.	9.1	>7.9	No
4.	6.8	≤7.9	No
5.	8.5	>7.9	Yes
6.	9.5	>7.9	Yes
7.	7.9	≤7.9	No
8.	9.1	>7.9	Yes
9.	8.8	>7.9	Yes

- So CGPA=7.9 is our best split point

How C4.5 Improves Upon ID3

Feature	ID3 Algorithm	C4.5 Algorithm (The Improvement)
Splitting Criteria	Uses **Information Gain**.	Uses **Gain Ratio** to prevent feature bias.
Data Types	Handles only **categorical/discrete** attributes.	Handles both **continuous and discrete** attributes by dynamically creating thresholds.
Missing Values	Fails or ignores records with **missing attribute values**.	**Handles missing values** by weighting the instances during evaluation.
Overfitting	Does not prune, leading to overly complex trees that **overfit** training data.	Uses **tree pruning** (e.g., error-based pruning) after creation to remove unhelpful branches.